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I am a Diploma-level student in the IIT Madras BS in Data Science program. I enjoy building real-world, impactful web apps using Python and modern JavaScript frameworks.

Description

This project aims to build a Placement Portal Application with three users: Admin, Student and Company, where company can create placement drive after approval from admin and student can apply for the role.

AI/LLM Used

ChatGPT was used primarily for debugging suggestions, code structuring tips, and UI improvements to some extend also used in backend. Which make roughly 10-15% use of LLMs according to Guideline Documents.

Technologies Used

Flask: Core backend framework

Flask-RESTful: For structuring APIs cleanly

Flask-JWT-Extended: For implementing secure authentication and role-based access

Flask-Caching: To improve performance by caching frequent API response

Flask-CORS: To handle cross-origin requests from Vue.js frontend

SQLAlchemy: ORM for DB modeling and querying

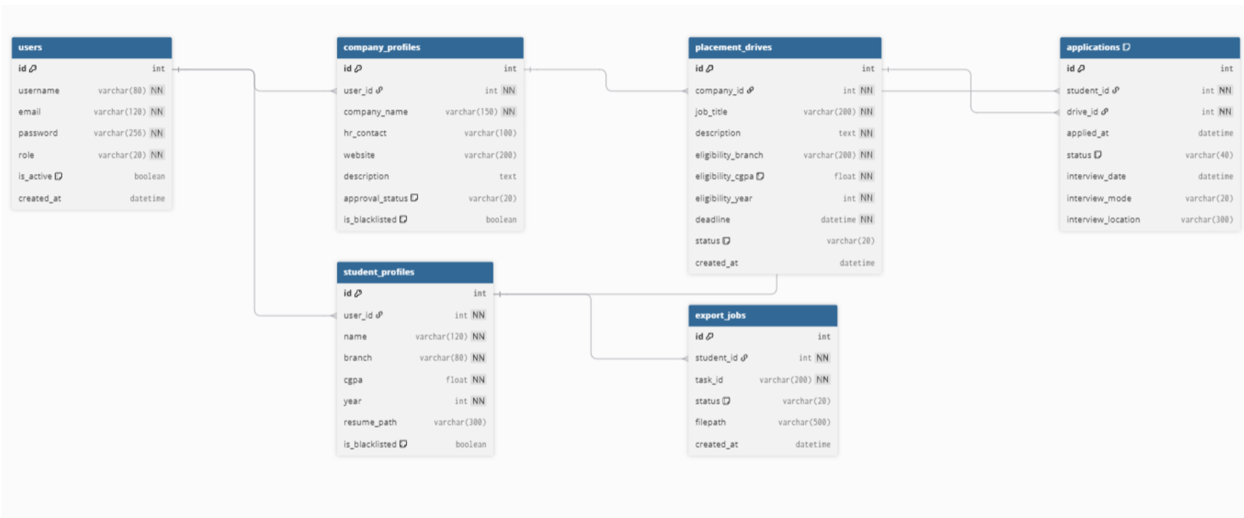
Vue.js: Frontend framework for building dynamic user/admin dashboards

Bootstrap: For responsive and aesthetic UI

Celery + Redis: For background tasks like scheduled reminders

Purpose: Each of these technologies was selected for clean API design, modular development, secure access, and scalable deployment. Vue + Bootstrap allowed for a lightweight but functional frontend interface.

DB Schema Design



API Design

APIs were created for the following modules:

- **Application:** Allows students to apply to placement drives with eligibility checks, resume upload, duplicate prevention, and authorized access to application details.

- **Authentication:** Handles user login and registration with JWT token generation, credential validation, role-based account creation, and profile initialization for students and companies.

- **Company:** Allows companies to manage their profile, view applicants for their placement drives, update student application status, and schedule interviews.

- **Drive:** Manages placement drives by listing approved drives, creating new drives, viewing drive details, retrieving company-specific drives, and searching drives or companies based on filters like job title, branch, CGPA, and year.

- **Student:** Allows students to manage their profile, upload and download resumes, view applications and history, export application data to CSV, check export status, and view applied placement drives.

- **Admin:** Allows administrators to manage the platform by viewing dashboard statistics, managing students, companies, drives, and applications, approving or rejecting companies and drives, searching users, and blacklisting or restoring user accounts.

- **Asynchronous task:** For reminders/reporting

All APIs are RESTful and implemented using Flask-RESTful.

Architecture and Features

The project follows a modular architecture with app.py as the entry point. API logic resides in the api/ folder, models in models.py. The frontend is built with Vue.js in the frontend/ folder.

Features Implemented

Default features include JWT-based authentication, role-based dashboards for admin, company, and students, placement drive creation and management by companies, and application submission and tracking for students. The system supports resume upload/download, eligibility-based application filtering, and interview scheduling by companies.

Additional features include admin dashboards with system statistics, API caching for improved performance (Flask-Caching), and background tasks such as exporting student application data to CSV using Celery and Redis.

Video

23f2001398_MAD2_Video